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Professional Summary

Computer and Systems Engineering student with hands-on experience across machine learning, data science, and backend development. Built and deployed production systems including an OT/ICS intrusion-prevention engine and a full-stack web platform. Skilled in Python, REST API design, ML model development, and scalable deployment. Seeking to apply strong engineering fundamentals to impactful, data-driven products.

Education

Bachelor of Science in Computer and Systems Engineering Ain Shams University

Expected Graduation: June 2026

Cumulative GPA: 3.0/4.0

Experience & Internships

- **Data Science Trainee**, DEPI (Digital Egypt Pioneers Initiative) *Dec. 2025 – June 2026*
 - Selected for the professional Data Science track under the Ministry of Communications and Information Technology (MCIT).
 - Practiced the end-to-end data science lifecycle: data acquisition, SQL querying, model deployment, and MLOps.
 - Used MLflow and Hugging Face for model versioning and experimentation with deep learning architectures.
- **Embedded System Essentials**, SIEMENS *Jul. 2025 – Sept. 2025*
 - Completed a comprehensive program covering Bootloader, RTOS, AUTOSAR, SDLC, and communication protocols.
 - Applied embedded software theory through hands-on labs and real-time, automotive-focused assignments.
- **AI Internship**, Samsung Innovation Campus (SIC) *Jul. 2024 – Oct. 2024*
 - Designed and implemented machine learning and deep learning models for computer vision and NLP tasks.
 - Built and optimized predictive models using TensorFlow, Keras, and Python for scalable performance.
 - Conducted exploratory data analysis and feature engineering to improve model accuracy; presented results to team leads.
- **AWS Public Cloud Summer Training**, National Telecommunication Institute (NTI) *Aug. 2023 – Oct. 2023*
 - Gained proficiency in AWS services, cloud security, architecture, and cost optimization.
 - Completed Red Hat System Administration 9.0: file management, permissions, and task automation.

Projects

• **Smart Shield – OT/ICS Intrusion Prevention System (GP)**

Python • Zeek • Redis • Docker • Flask • Linux • iptables • SIEM/Syslog

Production-ready Intrusion Prevention System that protects factory Operational Technology (OT) and Industrial Control System (ICS) networks. Deployed passively at the IT/OT boundary, it parses live traffic with Zeek, accumulates structured evidence, and safely blocks attackers while guaranteeing PLCs, HMIs, and SCADA servers are never disrupted.

My Role - OT/ICS Detection Modules:

- Designed and built the OT detection module identifying 20 attack categories across 6 industrial protocols (Modbus, S7Comm, DNP3, PROFINET, CIP, NTP/PTP).
- Implemented sliding-window flood detection and per-window alert deduplication to eliminate duplicate evidence under sustained attacks.
- Engineered the OT-safe blocking mechanism: a protected-device guard checked before every iptables rule, ensuring critical PLC/HMI/SCADA IPs are never auto-blocked.
- Authored the Zeek protocol script hooking Modbus/DNP3/S7Comm events and emitting custom notices (PLC mode switch, watchdog manipulation, valve/motor cycling).
- Produced structured Evidence objects integrated into the alert pipeline and exported to SIEM via Syslog (RFC-5424) and HTTP webhook.

• **EMS Full-Stack Content & Management Platform**

Python • FastAPI • PostgreSQL • Redis • SQLAlchemy • Alembic • JWT • Docker

Live production web platform with a content-management panel, contact workflow, and bilingual (Arabic/English) support. Built outside the university scope and currently serving real users at ems-eg.com (backend source kept private for security).

My Role - Backend Developer:

- Built the FastAPI backend with PostgreSQL and SQLAlchemy/Alembic migrations, exposing a documented REST API consumed by a React frontend.
- Implemented JWT authentication (access + refresh tokens) and a secured CMS for non-technical content updates.
- Configured CORS, environment-based settings, Redis caching, and SMTP email notifications for the contact workflow.
- Containerized the service with Docker and deployed to a managed cloud stack with PostgreSQL, automated migrations, and SSL.

• **FlipFlow: Inventory & Transaction Management System**

Python • Django • SQLite/MySQL • REST API • Threading • Git

Developed a distributed e-commerce platform with inventory tracking, user profiles, transaction history, and analytics. Led a team of 10 using agile methodology.

- Designed and implemented RESTful APIs using Django REST Framework.
- Built responsive frontend UI components using Bootstrap and jQuery.
- Implemented database replication logic using threading for fault tolerance.
- Integrated Chart.js for visualizing transaction trends and item popularity.
- Used Typst and Mermaid diagrams for documentation and planning visualization.
- Presented project architecture and time-plan using Gantt charts and timelines.

• **SteamMatch: Steam Video Games Recommendation System**

- Developed a content-based recommendation system for Steam video games using the Steam Video Games dataset.
- Utilized TF-IDF for the 'tags' column and Word2Vec for the 'description' column to extract relevant features.
- Implemented the recommendation model using machine learning techniques to suggest games based on user preferences.
- Worked with various tools including Python, scikit-learn, and TensorFlow for model building and performance optimization.

• **Employee Attrition Analysis for IBM**

Python • Pandas • Feature Engineering • Machine Learning

Analyzed HR dataset to identify factors influencing employee turnover and proposed data-driven retention strategies.

- Conducted exploratory data analysis (EDA) on a real-world IBM HR dataset containing 35 features.
- Applied Robust Scaler to handle outliers and used clustering and classification techniques.
- Built and evaluated multiple machine learning models using resampling techniques (e.g., SMOTE) due to class imbalance.
- Delivered actionable insights to help HR teams understand and mitigate employee turnover risks.

Technical Skills

- **Programming Languages:** C++, Python, Java, Embedded C, SQL, ARM Assembly
- **Backend & Web:** FastAPI, Django, REST API, PostgreSQL, MySQL, Redis, JWT, Docker, Git
- **Data Science & ML Tools:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow/Keras (basic)
- **Security & Systems:** Zeek, IDS/IPS concepts, Linux, iptables, SIEM/Syslog, OT/ICS protocols
- **Concepts:** OOP, Data Structures & Algorithms, Software Optimization, GUI Development, Agile

Languages

- **Arabic:** Native
- **English:** B1 (CEFR)

Interests & Activities

- **Sports:** Padel, Swimming
- **Skills:** Presentation, Team Collaboration, Problem Solving

REFERENCES

Available upon request